

From Mined Fixes to Gated Repair: Closed-Loop Defect Synthesis for CIRCT

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Abstract—Software bugs in hardware compilers like CIRCT risk injecting silent defects into physical chips. AI repair agents can fix known issues, but they are constrained by the slow trickle of human-reported bugs. We present a closed loop, built on the CHIA agent framework, that mines historical CIRCT commits to generate its own defect supply: a model learns the mistake pattern behind each of 187 past fixes and synthesises novel, bug-exposing inputs. The loop offloads building, judging, test reduction and deduplication entirely to deterministic tools, hands patch generation to CHIA’s existing repair chain, and reserves human effort exclusively for final filing approval. It restores the compiler’s internal assertions, compiled out of CHIA’s published image, in 536 of 555 objects, so defects become observable where none could fire before. Two registered campaigns then found fifteen new bugs, each reproduced, minimised, validated and unmatched against 2,652 issues, held for a named human to file, while a mutation baseline over 2,308 inputs found none; every model call of the project cost USD 189 at list price.

I. INTRODUCTION

CIRCT is an open-source compiler for hardware. A designer writes how a circuit should behave, in FIRRTL (the form the Chisel hardware language produces) or in SystemVerilog, and CIRCT lowers it to Verilog, which synthesis and layout tools turn into a chip. It is built on MLIR, the LLVM project’s compiler framework, and inherits LLVM’s way of testing: a regression test carries the command to run and the output to expect, run by a tool called `lit`. A bug that crashes CIRCT is an inconvenience, seen and worked around; one that makes it emit wrong Verilog is not: the output is well formed, passes the tools below it, and can reach fabrication unnoticed.

CHIA [1] is the agent framework this hackathon is built on. Its CIRCT issue solver chains six phases, from assessing an issue to reproducing, fixing, verifying against the whole test suite, repairing regressions and writing the pull request; on 16 CIRCT issues it fixed five, three of them merged.

Its supply is the constraint. Over the past year CIRCT received 82 bug-labelled issues and closed 65; what does not drain is the backlog of 101 open ones, at a median age of 911 days. The scarce resource is maintainer attention, and an agent that empties the queue faster creates none. What would help is a bug that arrives already minimised, already checked against what is known, already carrying a candidate fix, from a source that files nothing else. Past fixes are the unspent information: a commit that corrects one wrong assumption inside a pass usually leaves the same assumption at other sites, and CIRCT ships no fuzzing infrastructure to find them.

This paper describes a loop that manufactures its own supply and then withholds most of it.

II. THE LOOP

Fig. 1 names the stages and marks the ones the agent runs; tools decide everything else. One model serves every agent stage, Gemini 3.8 Flash through CHIA’s Vertex backend, and no stage can build a client for it unless one environment variable is set for that run, so a test or a dry run cannot reach the model.

Read the seed (1). A seed is one of 187 CIRCT commits from the last two years that look like bug fixes, with its diff and the test its author added or changed. The agent reads both, states in one sentence the root-cause class the fix corrected, and names other sites where the same assumption still stands.

Write probing inputs (2). The seed’s own test decides the language, because its command line names the tool through which the bug was reachable: `circt-opt`, `firtool` or the SystemVerilog front end. About 127 of the 187 seeds sit in code no FIRRTL input can reach, so a FIRRTL-only generator would miss most of the corpus. The agent reads the compiler’s source and writes each candidate through tools of its own, and runs no command.

Build (3). CIRCT compiles against one pinned commit of LLVM and MLIR, and Section III explains how the loop avoids building those from source. The build compiles in CIRCT’s assertions, the internal checks its own authors wrote, which stop the compiler the moment an assumption breaks.

Judge (4). An oracle is the rule that decides whether an input found a bug. The primary one admits no argument: the compiler crashed, or an assertion fired. A second runs the same design through CIRCT’s simulator and through Verilator, and treats disagreement as suspicious; it reports only, because the two disagree routinely over uninitialised and reset signals.

Reduce (5). A crashing input is usually large and mostly irrelevant to the crash. `circt-reduce`, which CIRCT ships, deletes parts of it and keeps each deletion while the failure survives; a textual reducer takes over where that tool dies.

Triage and report (6). Many failures are one failure, and deciding which are the same is mechanical: a fingerprint is the assertion text with its source location, or, for a crash, the symbolised frames left once the crash handler’s prologue is stripped. Survivors are matched against a local mirror of CIRCT’s open and closed issues and against commits made

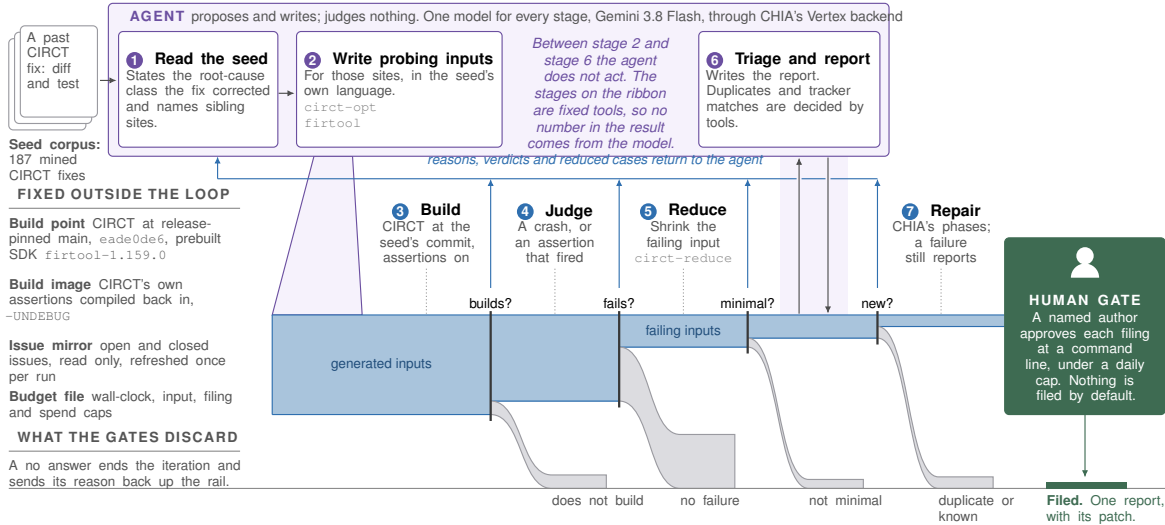


Fig. 1. Seeded CIRCT bug loop. The agent holds the violet band and acts three times only, at stages 1, 2 and 6; the blue ribbon below it is fixed apparatus, so no number in the result comes from the model. The ribbon is the candidates in flight, pinched at each gate by a yes-or-no question; the share answering no peels onto the floor. Widths are illustrative.

TABLE I
WHAT WE MEASURED ON THE IMAGE THE LOOP RUNS

Measurement	Result
<i>Building the past cheaply</i>	
Fix-shaped commits mined from 24 months	187
Of those, seeds with an exactly matching prebuilt SDK	171
Pin windows with no matching release, worst 16 days	16 of 63
<i>Letting the compiler complain</i>	
Compile commands carrying <code>-UNDEBUG</code>	778 of 778
Object files carrying the assertion handler again	536 of 555
CHIA's whole test gate, assertions on and then off	1,119 pass, 0 fail, identical
What the assertions cost: wall clock, bytes, per probe	18%, 25%, 4%
Tests <code>lit</code> discovers, once its source root is the SDK	1,390, 0 errors

since the build point. The agent writes the report, and no gate reads what it concluded.

Repair (7). CHIA's phase chain runs unaltered but for one branch selecting the backend the rest of the loop uses and an entry point taking a local report in place of a GitHub issue. Repair is restricted to crashes and assertion failures, where nobody has to argue about what the right answer was.

The gates. Between stages sit questions with two answers, the bars that pinch the ribbon in Fig. 1. A no ends the iteration and returns its reason to the agent with the verdicts and reduced cases. The last is a gate of four questions, each answered by a tool: does the failure reproduce in a fresh process on another worker, is the case minimal, is the input valid, and is it new. The third is three parse-and-verify commands, and a clean rejection makes the input the loop's own fault rather than the compiler's. Then the gate is a person: a named author approves every filing at a command line, under a cap on filings per day, and nothing is filed by default.

III. MEASURED PRECONDITIONS

The loop has to build CIRCT at arbitrary points of its history, and the compiler has to be able to complain. Neither held when we started; Table I is what we measured on the image the loop now runs.

Building the past cheaply. CIRCT records the exact LLVM commit it expects, its pin; a build against any other fails, and compiling that LLVM costs hours per seed. CIRCT also publishes an SDK of prebuilt LLVM and MLIR libraries carrying its own pin, and where the two pins match CIRCT rebuilds in minutes: 171 of the 187 seeds do.

The same mechanism fixes where the loop runs: not the newest commit of CIRCT's main branch, whose pin has no published release today, but release-pinned main, the newest commit whose pin has one, which for this image is 8 commits and 4.8 days behind the head. A long lag mechanically lowers the filing rate, because more candidates come back already fixed.

Letting the compiler complain. Optimised builds delete assertions by defining the macro `NDEBUG`, and CHIA's published CIRCT image is such a build: its binaries hold no reference to the assertion handler, so a loop whose oracle is a failed assertion would find nothing under it, whatever the agent wrote. One compiler flag, `-UNDEBUG`, puts them back, and the image the loop runs is that build at release-pinned main, with the flag in every compile command and the handler back in all but 19 of its 555 compiler objects, which assert nothing.

Whether the restored checks fire on CIRCT's own tests is what the oracle rests on, so we built the image twice, differing in that flag alone, and ran CHIA's whole test gate in both: the two runs pass the same 1,119 tests and fail none. No test passes with assertions compiled out and fails with them compiled in, so a probe that trips an assertion trips it because of what the probe does, and Table I prices the flag.

One defect had to be repaired first: at CIRCT commits after mid-2026 a test directory's configuration raises while `lit` discovers tests, and CHIA's exclusion of that directory hid it, so the gate discovered nothing. Pointing the build at the SDK as its MLIR source root, one flag, restores discovery, and every report names the flag.

Tested on real failures. The oracle and its fingerprints were calibrated on real output: six past CIRCT bugs, rebuilt at the commit before their fix and recorded with input, argument vector and error output, in three classes across two SDK releases, 14s to 601s of build each. Two defects in the fingerprinting rules appeared only there: the name normaliser erased every frame in an anonymous namespace, which is where CIRCT’s passes live, and the scope test read an inlined frame instead of the physical one, putting three of the four real bugs outside the repair stage’s reach. Both are fixed.

IV. CONTROLS

The experiment the loop embodies asks whether reading a past fix makes an agent better at finding new bugs than mutating the same tests at random. Two arms share the harness, oracles, reducer and gate, and only the generator differs: the seeded arm is stages 1 and 2, and the mutation arm rewrites the same seed tests syntactically without seeing the diff or the root-cause class, following Mut4All’s design for mutators synthesised by a language model [2]. Both arms spend the same budget, a fixed window of wall-clock seconds, one after the other on the same cluster at the same concurrency, and the artefact renders one headline: distinct bugs a CIRCT maintainer confirmed by labelling, commenting on or fixing them. Our oracle agreeing with itself produces a candidate, not a bug.

Pre-registration. Every threshold lives in one budget file, and a run is registered by an annotated git tag on the commit that lands that file; the loop refuses a file no tag names and a mutator set committed after the tag, so no threshold can be chosen after the fact. Safety caps bound damage rather than effort: 2,000 generated inputs per arm per day, 3 filings per day and 10 per run, 60 seconds and 4 GiB per probe, and a hard cap on model spending that stops both arms. Nothing is filed until a human has posted the method to the project’s forum, and issues marked as good first issues are refused.

Deduplication. Two candidates are one bug when a single primary fingerprint is equal. On a hand-labelled set of 22 candidate pairs the rule collides on 1 of the 11 distinct pairs and falsely merges 1 of the 11 duplicate pairs; both rates are printed beside the headline, since a collision inflates that count and a false merge deflates it.

Known issues. A mirror of the tracker answers whether a candidate is already reported. GitHub stops paginating at its hundredth page, so the mirror walks newest first, then oldest first, and unions the two: the whole tracker, 2,652 issues, 1,795 closed and 857 open, costs 198 requests and 200 seconds.

Contamination. CIRCT’s history sits in the training data of any model we can use, so the agent may reproduce a fix it memorised. Every candidate is screened against fixes made after its seed, at the level of the symbols the seed’s fix touched: that screen flags 77% of the corpus where the same screen at file level flags 96%. A real crash is a real crash, so the flag qualifies a candidate rather than excluding it.

V. RESULTS

Table II is two registered campaigns, each running the seeded arm as two shards in parallel. The first gave every arm two

TABLE II
BOTH CAMPAIGNS, FROM THE LOOP’S DATABASE. A SEEDED COLUMN IS ITS TWO SHARDS TOGETHER; THE MUTATION COLUMN IS ONE RUN PER CAMPAIGN, THE TWO AGREEING IN EVERY CELL

Measure	seeded 1	seeded 2	mutation
Seeds probed	9	43	96
Probes written	134	624	1,154
Output rejected by its own verifier	10	7	0
Assertion failures	5	54	0
Crashes	0	8	0
Candidates	15	69	0
Distinct new fingerprints	3	14	0
Through the gate	3	12	0
Filed	0	0	0
Model turns	55	191	0
Spend, USD at list price	44.5	122.8	0

hours and capped each job at USD 100, and was started three times, the first two stopped early for defects the live run exposed. The second changed three registered numbers and nothing else, the whole corpus instead of its head, a 41.7-hour window and a USD 175 cap, and we stopped it ourselves after 11.9 hours, well inside both. Every number is read from the loop’s database after the stored probes were re-judged under the two corrections below; no model turn was re-run.

Fifteen new bugs through the gate. The first campaign’s three came from one seed, the August fix for out-of-bounds `moore.extract` lowering, and sit beside it: an extract at the most negative 32-bit offset builds an array of zero elements and trips the assertion at `HWOps.cpp:1995`, while a dynamic extract and an extract of a reference lower to `comb.extract` and `llhd.sig.extract` with aggregate types, the compiler emitting IR its own verifier rejects. The second campaign’s twelve are spread across the compiler instead, and three are not assertions at all: a segmentation fault in `resolveClassStructBody` in `MooreToCore.cpp`, a second whose only named frame is `main` in `circt-opt.cpp`, and an abort in `ElaboratedObjectOp::verifySymbolUses` in `OMOps.cpp`. Each of the fifteen reproduced in a fresh process on another worker, was minimal at the reducer’s fixpoint, valid, and matched neither an issue nor a commit since the pin; the model wrote the reports, held for a named human. Two further new fingerprints did not survive it, one unreproducible on re-run and one rejected by the validity checker.

The oracle had to learn a class. Seventeen of the seeded arm’s rejected probes were the compiler’s fault, not the model’s: the input alone parses and verifies, and the diagnostic has the shape of MLIR’s generated verifier. Both conditions are needed, since a pass may complain legitimately about a valid input, such as two top-level modules. No rejection in either mutation run qualified: a mutated test that fails to parse is broken syntax, while the model’s inputs parse and exercise the lowering. That arm ran once per campaign, the two runs agreeing in every stored count, and across 2,308 mutants it raised no candidate to judge.

Dedup needed four corrections, each found on a real candidate. A post-pin commit counted as a fix if it touched the crash’s file, so an LLVM bump touching 87 files silenced the

first find; it now has to touch a frame symbol. A generic MLIR assertion text matched an open issue; it now also needs a CIRCT frame token. A verifier failure matched an issue on the op name alone; it now needs the constraint. And two candidates sharing a fingerprint were each the other’s duplicate, leaving thirteen without a representative. Every verdict is recomputed from stored evidence, so a corrected rule re-judges every candidate without a model turn.

Money. The first campaign spent USD 44.5 over 9 seeds and the second USD 122.8 over 43, about USD 4.9 and USD 2.9 a seed. Google’s cache served 68% of the first campaign’s 55.6 million input tokens and 60% of the second’s 148.7 million, so the real bill sits below the list price every figure here quotes: at the first campaign’s end the project had been billed USD 30 against USD 66 of list. Eight capped pilots preceded the campaigns for USD 13 in total.

Repair. Three attempts on the assertion. The first two never reached a patch, because the loop’s own script handed the tool two input files. With that fixed the chain reproduced the failure, wrote a one-line change, rebuilt CIRCT and ran its test gate, and was writing up when the 45-minute wall ended it; the change was a debug print. A fix is not claimed.

VI. RELATED WORK

SWE-bench [3] scores agents on GitHub issues and CHIA’s issue solver [1] applies the pattern to a hardware compiler; both draw from a queue, and we fill one. Fuzzers fill queues without repairing or withholding: CSmith [4] for C; for MLIR, MLIRod [5], Nüwa [6], DESIL [7] and FLEX [8]; our mutation arm follows Mut4All’s synthesised mutators [2]. AFuzz [9] is the origin of our method, an agent testing where else a reference bug’s cause could live; we apply it behind a gate. Delta debugging [10] is stage 5’s algorithm. Abstain-and-Validate [11], llvm-harness [12] and CodeRover-S [13] make a repair agent more selective; our gate sits downstream and discards by default.

VII. ARTEFACT AND LIMITATIONS

We release the loop as CHIA nodes and tools, so swapping the generator for the mutation arm is one node substituted. Its two halves meet at a frozen contract of seven record types and one function call, version 2.4, and the suite is 736 tests, 686 of which need no cluster, network or model. On CHIA’s cluster layer the loop’s five containers come up in 38 seconds and, with a recorded generator, drive 16 probes through both arms in 7.4 seconds. Four things go upstream to CHIA: the Dockerfile for the assertions-on image, its build workflow, three helpers for its CIRCT module, and the nineteen-line branch that lets its repair chain select our backend. All of it, with the corpus and every input, verdict, reduced case, transcript and gate decision, is at https://github.com/DevAbhinav-23/CHIA_Hackathon_2k26.

Limitations. Nothing has been filed at the time of writing; filing is the approver’s act and follows the forum post, and all fifteen wait on it. The second campaign covered 43 of the 187 seeds before we stopped it at 11.9 hours, so the corpus is not exhausted, and Vertex rate limiting rather than the cluster

bounded throughput at about 3.6 seeds an hour across two shards, one of which wrote no probe for two hours while its turns were still billed. Model time across every live run totals USD 189 at list price, and the repair chain’s own model use is not observable. The second oracle applies to 6 of the 187 seeds, which themselves rest on a subject-keyword filter with no diffs read to confirm each is a bug fix. The contamination screen misses a sibling fixed in a commit whose subject does not name it, and calibration by re-injecting known bugs is not possible on one prebuilt image, since no seed’s parent shares its LLVM pin. The image is built but not published, so a report names its content hash rather than a registry digest.

Scope. The loop does not repair wrong-output bugs, because the oracle that would find them is report-only, and it does not replace maintainers: it turns the human event from writing a bug report into reading one.

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